

On this page

Data Science Environment validation tool Configuration

- This Tool allows to automatically validate a Data Science environment. It creates the most common data science resources and interact with them to verify their expected behaviours (e.g. execution of Spark Jobs, gathering of enrichment data, db's connection). Regardless of the final state of the validation task, this tool must be able to clean up the environment, to leave it in a similar state to that before its execution

Validation Tasks

Order	Task	Description
1	data-set generation	Creates random data-sets in a temporary directory, according to the number of file-systems supported by the environment
2	data-set import	Copies the data-sets created in the previous task to the different file-systems supported by the environment
3	data source creation	Creates data sources on Pulse with the data-sets copied in the previous task. Then it executes data source analysis and validates their results
4	plan creation	Creates plans on Pulse with the data sources created in the previous task. Then it executes plan executions and validates their results
5	individual model creation	Creates two individual models, Feedzai (Random Forest) and H2O (Generalized Linear Modeling) using cross validation. Then it executes another model evaluation with feature importance and validates their results
6	hyperparameter tuning creation	Creates a Hyperparameter Tuning with a Feedzai (Random Forest) and H2O (XGBoost) and validates that they execute successfully in the cluster.
7	rule creation	Creates two/three rules and publishes the application. Then it creates a snapshot evaluation and validates its result
8	sandbox creation	Creates a sandbox evaluation with the models and snapshot created in the previous tasks and validates its result. By default it used the default outcome combinator but it is possible to configure it.
9	automl creation	Creates an AutoML and validates that it finds a best model successfully.
10	jupyterlab integration	Checks whether JupyterLab is configured. If so, then tries to start it via Pulse.
11	scheduled metrics job creation	Creates a Scheduled Metrics Job (SMJ) on Pulse using the resources created in the previous tasks. Then it executes a job execution and validates their results

Ansible variables

Variable	Type	Default Value	Description	task
tool_datascience_plan_enabled	boolean	true	Whether this validation should be skipped or not.	plan creation
tool_datascience_plan_deployable	boolean	true	Whether this validation should be skipped or not.	plan creation
tool_datascience_model_enabled	boolean	true	Whether this validation should be skipped or not.	individual model creation
tool_datascience_hyperparameter_enabled	boolean	true	Whether this validation should be skipped or not.	hyperparameter tuning creation

Variable	Type	Default Value	Description	task
tool_datascience_rule_enabled	boolean	true	Whether this validation should be skipped or not.	rule creation
tool_datascience_sandbox_enabled	boolean	true	Whether this validation should be skipped or not.	sandbox creation
tool_datascience_jupyterlab_enabled	boolean	false	Whether this validation should be skipped or not.	jupyterlab integration
tool_datascience_automl_enabled	boolean	false	Whether this validation should be skipped or not.	automl creation
tool_rte_scheduledmetricsjob_enabled	boolean	true	Whether this validation should be skipped or not.	scheduled metrics job creation
tool_cleanup_resources_enable	boolean	true	Whether this validation should be skipped or not.	
tool_failure_cleanup_enable	boolean	true	Whether this validation should be skipped or not.	

As an example, Data Science Environment validation tool tool default configuration includes the following execution task:

```
- name: "[900] Run Data science environment validation tool"
  become: yes
  become_user: "{{ pulse_validation_tool_unix_user }}"
  shell: "{{ run_command }}"
  when: enable_run_tool
```

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